ANTIMA MODAK

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Profile

Software Engineer with 3 years of experience building backend systems, APIs and integrations for SaaS and healthcare clients, and an MSc in Applied AI from the University of Warwick. Builds practical AI products (LLM/RAG applications, explainable ML) and works directly with clients to turn requirements into shipped software.

Key Skills

Languages: Python, JavaScript, SQL, C++
Backend & APIs: REST APIs, FastAPI, Flask, Node.js, JWT authentication, AWS Cognito, Webhooks
AI & Data: LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, XGBoost, NLP, SHAP, AI coding tools
Databases: PostgreSQL, MongoDB, MySQL
Cloud & Tools: AWS Lambda, Docker, Git-based CI/CD (Vercel, Render), React, n8n workflow automation

Work Experience

Warwick Business School, University of Warwick – United Kingdom **March 2026 – Present**
Applied AI Analyst (Developer)

- Developed backend services and REST APIs in Python and Flask (Supabase) for a healthcare application tested by 10+ internal users, supporting secure data exchange and cloud deployment.
- Led end-to-end development as the sole developer, from stakeholder requirements through implementation, testing, deployment and iteration, working directly with researchers, physiotherapists and domain experts.
- Set up and maintained Git-based CI/CD deployment pipelines through Vercel, covering build automation, deployment checks and release hygiene.
- Accelerated prototyping and debugging with AI-assisted development tools, while owning all architectural decisions and reviewing AI-generated code for correctness.

TrueLoyal (Formerly Zinrelo) – Pune, India **June 2023 – September 2025**
Full Stack Developer

- Served as primary technical contact for 10+ external clients, gathering requirements and delivering customised CRM and loyalty integrations end to end.
- Developed and deployed 10+ AWS Lambda functions in Node.js through automated build and release pipelines, reducing manual workload by over 40%.
- Integrated AWS Cognito for secure user login and signup, reducing authentication-related support tickets by 25%.
- Improved API response time and data consistency by 15% by optimising backend queries on PostgreSQL and MongoDB.
- Improved Shopify theme app extension performance, reducing page load time by 35%.
- Built 20+ automated workflows in n8n orchestrating multi-step business processes, improving internal efficiency by 30%.
- Accelerated development, debugging and code review across client projects using AI coding assistants (Claude in VS Code).

Projects

Enterprise AI Knowledge Platform (LLM/RAG) **July 2026**

- Built a full-stack RAG platform designed for hundreds to thousands of enterprise documents, with source citations, audit logging and role-based access control.
- Implemented document ingestion, chunking, embeddings and vector similarity search (pgvector), with LLM answer generation using Ollama (local) and Gemini (cloud).
- Developed JWT authentication, query history and audit logging for enterprise-style governance; React frontend with FastAPI and PostgreSQL backend.
- Containerised with Docker and deployed on Render and Vercel via Git-based CI/CD.

Explainable AI for Product Return Prediction (MSc Dissertation) **June – July 2026**

- Built a 10-phase pipeline classifying product-return root causes at 97.1% accuracy (XGBoost; Macro F1 0.964, AUC 0.999).
- Applied an NLP pipeline (VADER, LDA, TF-IDF) to 50,000 reviews and added SHAP so every prediction is auditable.
- Engineered a composite risk-scoring framework quantifying £308,000 in annual return cost exposure, with a local LLM (Ollama) generating stakeholder recommendations in a Streamlit dashboard.

VTap: AI-Powered Payment Optimisation Tool **March 2026**

- Reached the Top 4 (of 26 teams) at a Google London hackathon with a payment tool that recommends which card to use based on available balance and best merchant rewards; presented to judges from Google and NatWest.

Education

University of Warwick, WMG – United Kingdom **September 2025 – September 2026**
MSc Applied Artificial Intelligence
Dissertation: An Explainable AI Framework Integrating Customer Feedback and Transactional Data for Product Return Prediction in Supply Chains
Modules: AI & Deep Learning, Programming for AI, Applied Statistics for AI, Data Mining, AI for Industry, Ethical AI

Vishwakarma Institute of Technology – India **August 2019 – June 2023**
B.Tech in Computer Engineering – 9.2/10 CGPA

Leadership

Student Ambassador, University of Warwick (WMG) **September 2025 – Present**

- Represented the MSc programme at open days, delivering presentations and Q&A sessions to prospective students.